

# ISAAC ADJEI

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## PROFILE

Electronic Engineering and Computer Science student at Aston University (Predicted First Class), specialising in embedded systems and hardware-software integration. Built real-world systems from circuit and firmware level through to secure full-stack applications, including embedded systems, analogue circuit designs and production-style and scalable web platforms. Seeking internships, placements and professional roles or opportunities in engineering and technology.

## EDUCATION

### Aston University, Birmingham, United Kingdom

Oct 2024 – Jul 2028

BEng (Hons), Electronic Engineering and Computer Science · **Predicted: First Class**

Top 40 Finalist – Black Heritage Undergraduate of the Year Award 2026 (TargetJobs & Sky)

**Relevant Modules:** Internet Applications and Databases, Foundations of AI and Data Science, Introductory Programming for Digital Science (Python), Electronics 1, Electronics 2, Introductory Mathematics for Engineering, Power Skills

- Developing strong foundations in software engineering, electronics and applied mathematics
- Building practical project experience across embedded systems, AI/data and full-stack development

### Stanmore College, London, United Kingdom

Sep 2022 – Jul 2024

Pearson BTEC Level 3 National Extended Diploma in Engineering · **D\*DD (Distinction\*, Distinction, Distinction)**

Best and Most Hardworking Student Award

**Relevant Modules:** Engineering Product Design and Manufacture, Microcontroller Systems for Engineers, Engineering Principles, Specialist Engineering Project, CAD in Engineering, Electronic Devices and Circuits, Calculus to Solve Engineering Problems, Electronic Measurement and Testing of Circuits

- Completed intensive hands-on training in engineering design, testing and technical documentation
- Applied microcontroller, electronics and CAD skills through practical coursework and projects

## TECHNICAL SKILLS

**Languages:** C, C++, Python, TypeScript, JavaScript, Java, PHP, SQL, MATLAB

**Embedded & Hardware:** Arduino, STM32, ESP32, Raspberry Pi, AVR, KiCad, Proteus, PCB Layout, Circuit Design, Oscilloscope, I2C / UART / SPI

**Web:** Next.js, React, Node.js, Tailwind CSS, PHP 8.2, MySQL, Apache

**AI / ML:** TensorFlow, PyTorch, NumPy, Pandas, Scikit-Learn, OpenCV

**Cloud & DevOps:** AWS, Azure, Docker, Vercel, Cloudflare, GitHub Actions

**Professional:** Problem Solving, Critical Thinking, Communication, Team Collaboration, Leadership, Time Management, Adaptability, Attention to Detail

## PROJECTS

### PHAEMOS: Smart Maintenance Platform

2026, ongoing

FastAPI, Python, Next.js 14, PostgreSQL, Redis, scikit-learn, Docker, ESP32, Arduino, STM32

[GitHub ↗](#) | [Website ↗](#)

- End-to-end IoT platform: ESP32, Arduino and STM32 hardware nodes POST sensor telemetry to a FastAPI backend with PostgreSQL and Redis, processed in **under 200ms per request**
- Unsupervised Isolation Forest anomaly detection scores every telemetry reading; above 0.7 triggers an alert and auto-generates a maintenance ticket, above 0.85 attaches a diagnostic recommendation
- Next.js live dashboard polls every 5 seconds with Recharts line charts; anomalous readings highlighted in real time
- **JWT auth with role-based access control** (Admin, Technician, Viewer) enforced at API and UI level with a full **immutable audit log**
- Active build targeting an industry-scale predictive maintenance product; architecture and API designed for production deployment from the outset

### Two-Stage Audio Amplifier

2026

Analogue Design, PCB Design, KiCad, Proteus, Electronics

[Portfolio ↗](#)

- Designed and built a two-stage amplifier using a TL071 active band-pass filter and OPA551 output buffer
- Delivered a **full engineering workflow: simulation, breadboard validation, PCB layout and bench testing**
- Validated performance with measured output of **2.980 Vpp against a 3 Vpp design target**

### 4x4x4 NeoPixel LED Cube

2025

Arduino, C++, WS2812B, Embedded Systems

[GitHub ↗](#) | [Portfolio ↗](#)

- Built a 64-LED interactive display with a **non-blocking state machine architecture** using millis()
- Implemented four animation modes, ambient-light adaptive brightness and debounced user controls
- Integrated diagnostics and electrical protection for reliable real-world operation
- Designed to demonstrate scalable embedded system architecture and responsive real-time control without blocking delays

## AstonCV – Full-Stack CV Database

2026

PHP 8.2, MySQL, CSS, Apache, Cloudflare

[GitHub ↗](#) | [Portfolio ↗](#)

- Built a full-stack platform from scratch with no frameworks, including authentication and CV management
- Implemented **11 security controls** including **bcrypt hashing**, **CSRF tokens**, **PDO prepared statements** and **brute-force protection**
- **Deployed live** with searchable profiles, sorting/filtering and server-side PDF export
- Engineered as a secure, scalable system applying real-world backend security practices and deployment workflows

## CNC Milling Machine Control System

2024

Arduino, C++, Safety Systems, State Machines

[Portfolio ↗](#)

- Implemented a **safety-focused controller** using **finite state machine logic** and strict transition validation
- Integrated door interlocks, **emergency stop handling**, timed cooldown and LCD status feedback for safe operation

## git-unlocked – Open Source Git Course

2026

Git, GitHub, GitLab, Markdown, Open Source

[GitHub ↗](#) | [Portfolio ↗](#)

- Created a structured learning resource with **170+ topic files** from fundamentals to advanced workflows
- Covered Git and major hosting platforms with practical cross-platform guidance for real teams
- Maintained contributor workflow with issue templates, PR standards and CI checks

## EXPERIENCE

### Student Representative

**Aston Students' Union, Birmingham, UK**

Student advocacy, stakeholder communication, academic liaison

Sep 2025 – Present

- Collect and communicate student feedback to academic staff through SSLC, Senate and Council meetings
- Delivered improvements to learning delivery and the wider course experience

### Engineering Operations & Maintenance Planning (Virtual Experience)

**British Airways via Forage, Remote**

Maintenance planning, risk analysis, reporting

Oct 2025

- Analysed A320 maintenance schedules, diagnosed component faults and produced structured WREQ engineering reports
- Developed material forecasts and maintenance planning reports for six aircraft with risk-mitigation recommendations

### Smart Mobility & Digital Tech (Virtual Work Experience)

**Yunex Traffic via Springpod, Remote**

Intelligent transport systems, sensor integration, data-driven urban mobility

Aug 2025

- Analysed intelligent transport systems and digital solutions to evaluate their impact on air quality and urban mobility efficiency
- Delivered technical insight outputs based on real-time pollution sensing and smart traffic workflows

### Consular Intern

**Ghana High Commission, London, UK**

Consular operations, document processing, client support

Jul 2024 – Aug 2024

- Managed passport and visa processing workflows, including document verification and client-facing support
- Implemented document verification and administrative coordination while improving communication and public-service professionalism

### Diplomatic Administrative & Estates Intern

**Ghana High Commission, London, UK**

Facilities coordination, document handling, office operations

Mar 2024

- Managed administration section workflows including estates management, document handling and general office operations
- Implemented facilities coordination and internal logistics, strengthening professional communication and organisational efficiency

### Junior Apprentice HVAC Technician

**Massive Refrigeration Services, Accra, Ghana**

Field servicing, diagnostics, installation support

Jul 2019 – Jul 2021

- Serviced, installed and maintained 50+ air conditioning units using standard diagnostic tools

## VOLUNTEERING

### Student Judge – National Emerging Talent Awards 2026

Feb 2026

targetjobs UK (Group GTI)

- Evaluated employer submissions for Best Placement or Internship Programme using qualitative and scoring criteria
- Commended for a thorough, timely and high-quality approach

### Fundraising Volunteer – 10 Days of 5K Challenge

Feb – Mar 2026

Cancer Research UK

- Participated in the Cancer Research UK 10 Days of 5K Challenge and completed 10 x 5km runs (more than 50km total) across March, demonstrating consistency and discipline
- Set up and managed an online fundraising page, contributing to a wider campaign that raised over £797,424.64, with an additional £159,224.14 through Gift Aid
- Raised funds through outreach and personal network engagement while promoting awareness of cancer research initiatives

## SPOKEN LANGUAGES

**English** (Full professional), **Twi & Ga** (Full Native), **French & Spanish** (Elementary)